

Lab-05: Clean Up and Push to Remote Repository – Implementation

Purpose

The purpose of this lab is to learn how to synchronize the local Git repository with the remote GitHub repository by checking the repository status, listing available branches, pulling the latest changes from the remote repository, and pushing local commits to the remote repository.

Implementation

Step 1: Navigate to the Existing Repository

```
cd Desktop
```

```
cd GitDemo
```

Step 2: Verify the Repository is in a Clean State

```
git status
```

Expected Output:

On branch main

Your branch is up to date with 'origin/main'.

```
nothing to commit, working tree clean
```

Step 3: List All Available Branches

Display all local and remote branches.

```
git branch -a
```

Example Output:

```
* main
remotes/origin/main
```

Step 4: Pull the Latest Changes from the Remote Repository

Pull the latest updates from GitHub.

```
git pull origin main
```

Expected Output:

Already up to date.

If there are new changes on GitHub, Git downloads and merges them automatically.

Step 5: Push Local Changes to the Remote Repository

Push all pending local commits to GitHub.

```
git push origin main
```

Expected Output:

Everything up-to-date

or

Enumerating objects...

Counting objects...

Writing objects...

To <https://github.com/Adharshinivankayala/GitDemo.git>

Step 6: Verify the Changes on GitHub

Open your GitHub repository:

<https://github.com/Adharshinivankayala/GitDemo>

Refresh the page and verify that:

- All committed files are present.
- Recent commits are visible under **Commits**.
- The repository is synchronized with your local project.

Commands Executed

```
cd Desktop
```

```
cd GitDemo
```

```
git status
```

```
git branch -a
```

```
git pull origin main
```

```
git push origin main
```

What Was Implemented

- Opened the existing Git repository.
- Verified that the repository was in a clean state.
- Listed all local and remote branches.
- Pulled the latest changes from the remote GitHub repository.
- Pushed all pending local commits to the remote repository.

- Verified that the repository on GitHub reflected the latest changes.

Result

The Clean Up and Push to Remote Repository Hands-On Lab was successfully completed. The local repository was synchronized with the remote GitHub repository by verifying the repository status, listing available branches, pulling the latest changes from GitHub, and pushing local commits. The remote repository was updated successfully, ensuring both local and remote repositories remained synchronized.

```
Command Prompt
C:\Users\vanka\Desktop\GitDemo>git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

C:\Users\vanka\Desktop\GitDemo>dir
Volume in drive C is Windows
Volume Serial Number is 106D-CD90

Directory of C:\Users\vanka\Desktop\GitDemo

04-07-2026  14:24    <DIR>          .
04-07-2026  11:58    <DIR>          ..
04-07-2026  14:10                23 .gitignore
04-07-2026  12:18                22 application.log
04-07-2026  12:41                43 branchfile.txt
04-07-2026  14:09                91 hello.xml
04-07-2026  12:18    <DIR>          log
04-07-2026  11:58                19 welcome.txt
               5 File(s)                198 bytes
               3 Dir(s)  256,746,233,856 bytes free

C:\Users\vanka\Desktop\GitDemo>git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

C:\Users\vanka\Desktop\GitDemo>git branch -a
GitWork
* main
  remotes/origin/main

C:\Users\vanka\Desktop\GitDemo>git pull origin main
From https://github.com/AdharshiniVankayala/GitDemo
* Branch      main      -> FETCH_HEAD
Already up to date.

C:\Users\vanka\Desktop\GitDemo>git push origin main
Everything up-to-date
```